

Coandă-Effect VTOL UAV Prototype, 2008–2009

During 2008–2009, I constructed an experimental saucer-shaped vertical takeoff and landing UAV based on the published GFS flying-saucer concept. The aircraft used an electric motor to direct airflow over its curved upper surface, applying the Coandă effect to generate lift for vertical takeoff, hovering, and landing without a large exposed rotor.

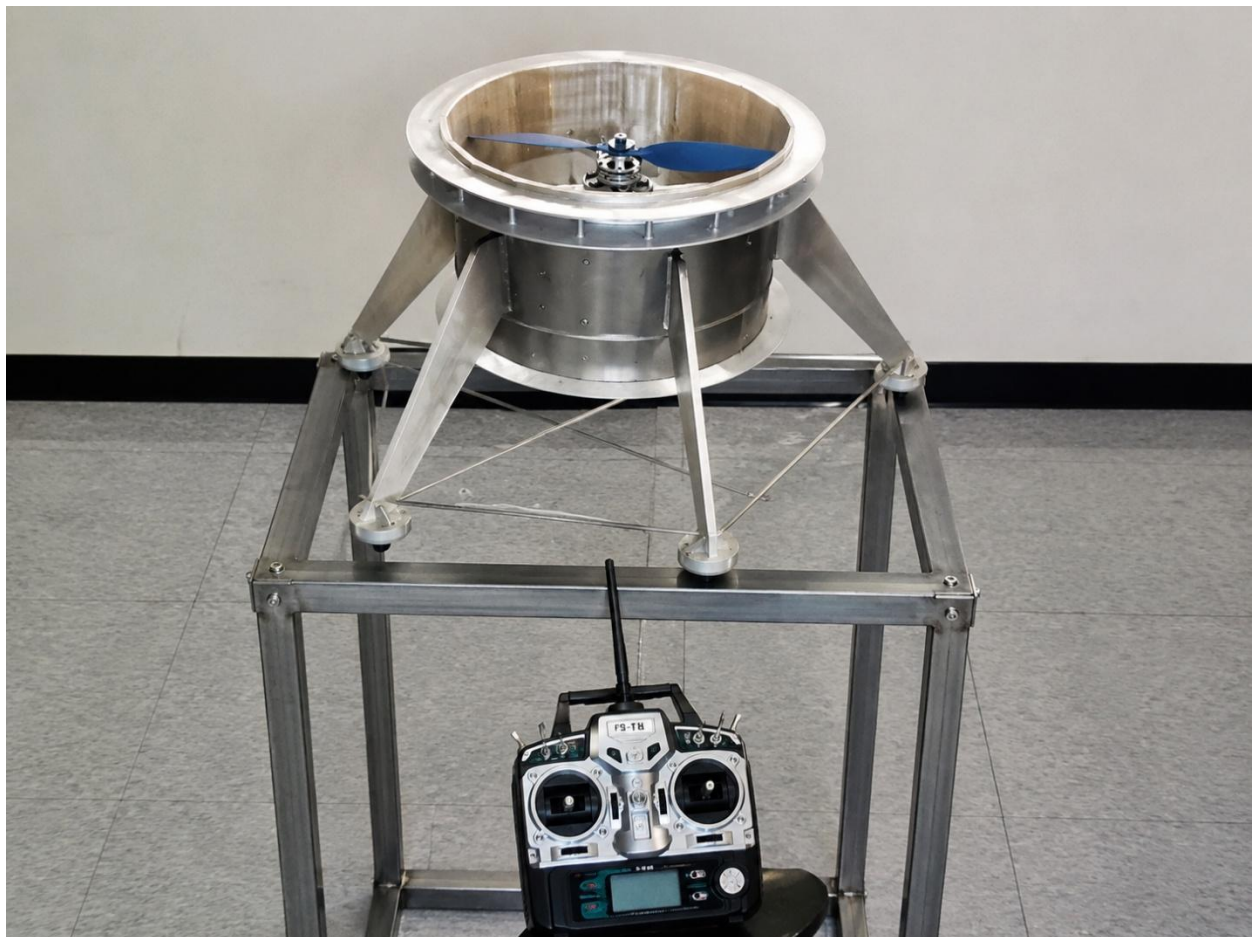
The initial structure was copied from the GFS configuration and then substantially redesigned. I reduced the airframe weight, revised the body structure and divider-ribbon arrangement, and improved the aerodynamic and propulsion efficiency. These modifications made my prototype lighter than the original design and increased its practical flying time.



Single-Ducted-Fan VTOL Prototype | 2009–2011

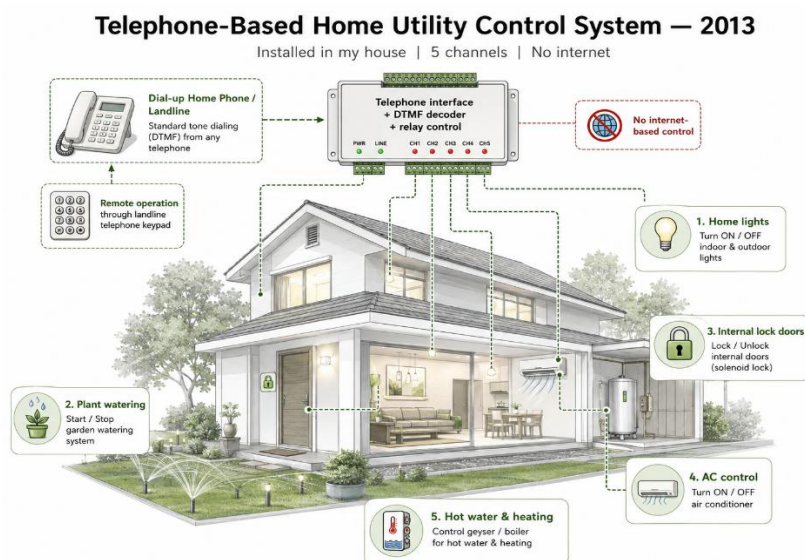
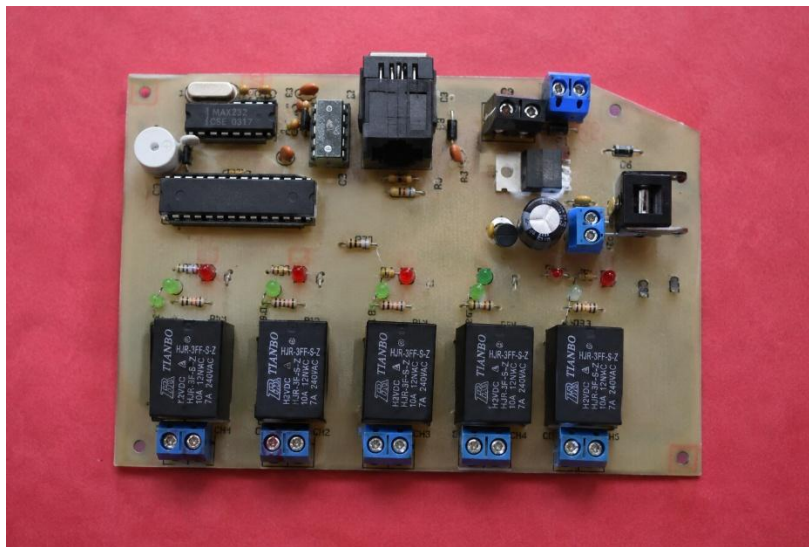
I designed and developed a remotely controlled vertical takeoff and landing prototype based on a single ducted-fan propulsion system. The project included fabrication of the supporting structure, integration of the electric motor and rotor assembly, installation of the protective duct, and preliminary ground testing using a radio-control system. The engineering investigation focused on hover operation, structural stability, thrust generation, rotor-duct interaction, and blade-tip clearance. Tip clearance is a critical design parameter because leakage flow near the blade tips can reduce aerodynamic efficiency and overall thrust performance in ducted-fan VTOL systems.

The prototype demonstrated a practical early-stage concept for a compact, single-duct VTOL uncrewed aerial vehicle.



Telephone-Based Home Utility Control System — 2013

In 2013, I designed, fabricated, and installed a five-channel home utility control module that operated through a conventional landline telephone, without an internet connection. The system used a telephone line interface and a DTMF tone decoder to interpret commands entered via a remote telephone keypad. A control circuit then activated independent output relays for household lighting, plant-watering equipment, internal door locks, air conditioning, and hot-water and heating systems. This architecture is consistent with telephone-based access and automation systems in which keypad commands remotely operate relay-controlled devices.



Wind Turbine Project, 2016

This project developed a small experimental vertical-axis wind turbine by mounting a metal rooftop turbine ventilator on a fabricated steel support frame. The turbine is designed to capture wind from different horizontal directions and convert the airflow into rotational mechanical motion. Based on the reference design, the rotating shaft can be connected to a low-friction bearing and a small DC motor or generator to produce electrical energy. The prototype was intended to demonstrate the basic principles of vertical-axis wind-energy conversion, turbine rotation, mechanical transmission, and small-scale renewable electricity generation.

